# Detection of Fallen Electric Cables on Roads and Streets in Inner Cities for Autonomous Vehicles and Bikes

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**Urban Fallen Cable Detection Dataset (UFCDD): A Visual Benchmark for Smart Mobility in Pakistan**

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## Abstract:

It has been a few years now that technology has focused on making tasks easier and faster for humans. One important invention in the past two centuries is the car which is now known as the key player in modern transportation, the autonomous vehicle. Research on how to make these vehicles easier to maneuver and safer does not stop. Considering how hard urban road technology makes it to detect and react to fallen cables and even though it matters, there are few resources exploring this in the literature. It is clear to all that running over such cables can be catastrophic for both vehicles and people, as it can cause their parts to fail, prevent normal movement of vehicles or even endanger lives. Specific images focused on this issue in Pakistani urban areas have been arranged, so these are available to support AI systems that search for dangerous cables. The objective is to give self-driving vehicles increased smartness in urban environments.


## Introduction

Technological progress, of course, has influenced the way we live and interact with others over time. Within this evolution, autonomous vehicles represent a promising solution for smart and efficient mobility. However, with more reliable and secure self-driving vehicles emerging around the world, researchers are now aiming to develop vehicles capable of addressing real-world problems. Among these critical issues, fallen electric and communication cables on roads, particularly in certain urban environments in Pakistan, stand out. Standard datasets and detection systems have largely overlooked these obstacles. They can cause accidents, damage vehicles, and pose safety risks if left undetected. To address this gap, our project aims to create a dedicated dataset to facilitate the vision-based detection of such hazards, in preparation for safer autonomous driving.

## Research Problem:

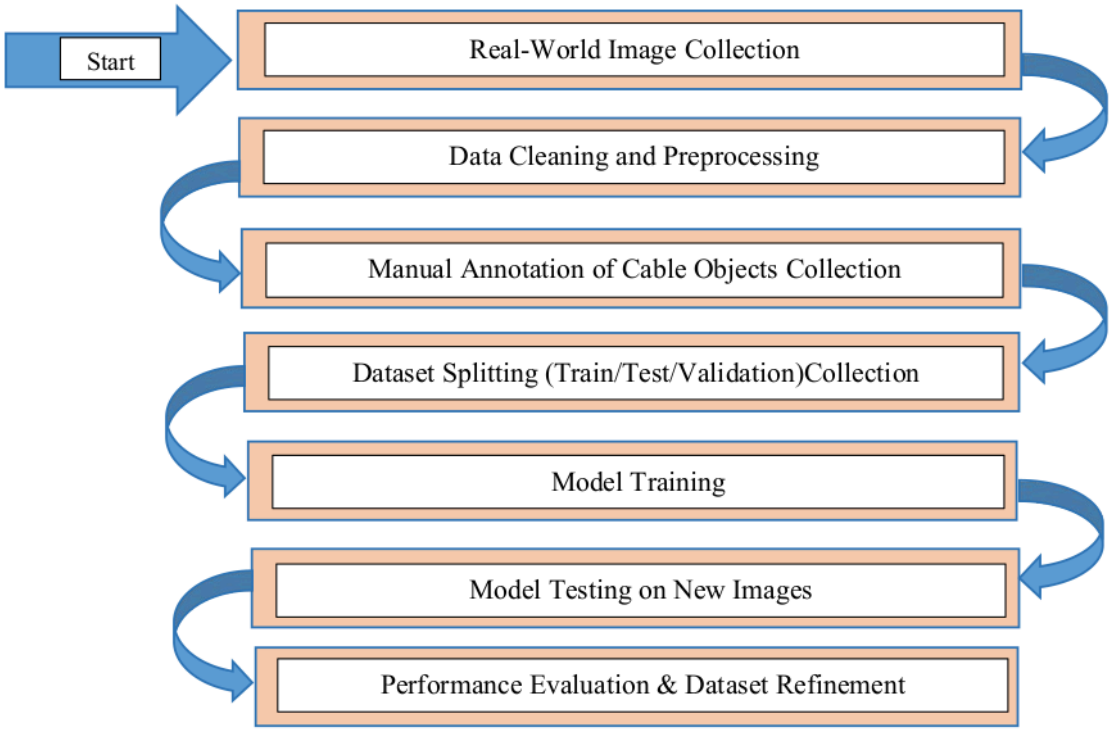
Things have moved so much forward that it's a sure bet you'll not use traditional methods anymore. Inventions and new technologies make life simpler and take less effort. Autonomous vehicle technology is developing very rapidly. Researchers and engineers all over the world are generating solutions to improve how these vehicles drive.

However, a critical challenge has emerged in the context of autonomous driving—these vehicles often fail to detect fallen power and communication cables (such as electric lines) lying on roads.Because of the current climate and the effects of climate change, dust storms are happening more than before which has led to more electrical cables breaking and falling onto roads. Issues arise since this shortcoming makes it dangerous for autonomous vehicle when accidents are likely to happen. As well as slowing vehicle travel, damage to the vehicle and a higher risk of system issues or incidents may appear.

## Proposed Solution

So, we are proposing to develop a specific dataset that can catch fallen electric cables on the streets of cities. We will get this dataset by collecting it under different weather examples and at various times of the day, resulting in a variety of actual case studies.

The collected data will be used to train a Convolutional Neural Network (CNN) model. For training, we will define two labels: "Safe" and "Unsafe." These labels will indicate which cables are intact and pose no risk (safe), and which cables are fallen or hazardous (unsafe). The CNN will receive these extra categories as input which will make detecting cables more accurate in autonomous vehicles.




**FlowChat Diagram**

## Literature Review

The detection of power lines and transmission towers (collectively referred to as pylons) is a critical task in improving the safety of low-flying aircraft, including manned and unmanned aerial vehicles (UAVs). Modern studies indicate that people are using deep learning frameworks instead of traditional methods for improved and more effective detection.

A novel vision-based system proposed by [Jakub Gwizdała(s), 2025] employs a hybrid deep learning framework combining ConvNeXt and MobileNetV4 networks for simultaneous detection of cables and pylons. It uses transfer learning and creates a new loss function to make sure power cables are more easily separated from the image. Having two networks in this system greatly boosts detection performance by taking care of both the overall reference areas within a scene and the more detailed boundaries of objects. The importance of such detection systems is underscored by Federal Aviation Administration (FAA) data, which reported an annual average of 76 aircraft-power line collisions between 1997 and 2010, with a fatality rate of 30% (FAA, 2010).[i]

Early research in this field focused on extracting power lines using techniques like the Radon transform, grouping strategies, and Kalman filtering to detect and reconstruct line structures in complex aerial backgrounds (Zhang et al., 2013). These tracking techniques worked fine in controlled spaces, although they had trouble handling background interference and troubles seeing cables clearly. Regular line detection methods could not compare in terms of performance because the radar system was able to see through broken structures and identify them clearly due to Radon transform, along with data grouping and Kalman filters.[ii]

In the past few years, research has used deep learning to fix the issues in classical approaches. A convolutional neural network (CNN)-based method was proposed to detect cables in UAV imagery by encoding them as sets of evenly distributed key points rather than entire lines (Liu et al., 2020). Because of this simplification, it became easier to detect collisions and more accurate to create the curves with implicit functions. Certainly, experts also created new ways to label and improve data to help the model remain effective in different conditions.[iii]

To improve real-time monitoring and detection, some researchers have integrated Internet of Things (IoT) sensors with UAV imaging systems. A hybrid deep learning model combining CNNs and Transformers was introduced to detect anomalies in power transmission infrastructure (Kumar et al., 2022). On the transmission lines, IoT sensors sensed any unusual event and this triggered drones to gather detailed pictures for inspection. CNN identified local features in the video and the Transformer understood the overall connections which boosted precise detection and lowered the number of errors. Large-scale detection of anomalies in power grids was possible using the model with high efficiency and scaled-up ability.[iv]

Moreover, sample matching and post-clustering techniques have been developed to further refine detection results and minimize false positives (Wang et al., 2021). Research revealed that these

methods are more efficient than the conventional edge and line detectors, so CNN is a reliable technique for ensuring safety in UAV-based inspection and navigation tasks.[v]

## Objectives

1. A range of fallen power and communication cables will come from different urban points in Pakistan.
2. Data will be gathered in the early morning, afternoon, evening just as in real life.
3. The data will be manually tagged with fallen cables so they are easy to spot.
4. A public dataset will be built to help train and test systems that detect hazards in autonomous vehicles.

## Scope

The dataset will cover multiple area in Pakistan (Mirpur) to account for geographical and infrastructural diversity.

1 Environmental conditions will include:

i Clear daylight

ii Overcast and cloudy weather

2 We will put our data into the time zones of each client as follows:

i Early morning (dawn)

ii Late morning

iii Afternoon

iv Evening (dusk)

## Conclusion

In this work, we study a serious, but overlooked danger for autonomous vehicles in cities which is caused by fallen power cables. The goal is to supply a suitable collection of images for training computer vision to find real-world obstructives in Pakistan, with this study having gathered pictures from actual local situations.

With the same processes, researchers can focus on automotive safety and also explore new ways to create better tech. At the same time, the dataset has high value outside of autonomous cars: it can be applied to robotics fields including drone navigation, delivery robotics and surveillance systems.

Generally, the using this data will improve how smoothly and safely autonomous vehicles operate in roadways with several lanes. Also, this knowledge can be used in many other smart mobility fields and has an important part in making transportation systems smarter in developing countries.

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[i] FAA. (2010). *Analysis of NTSB aviation accident data 1997–2009*. Federal Aviation Administration

[ii] Zhang, Y., Li, Z., & Zhou, H. (2013). Automatic power line detection in aerial images using Radon transform and Kalman filter. *Journal of Applied Remote Sensing*, *7*(1), 073577.

[iii] Liu, H., Zhao, Y., & Sun, C. (2020). Key-point encoding for thin object detection in UAV imagery. *ACM Transactions on Intelligent Systems and Technology (TIST)*, *11*(4), 1–19.

[iv] Kumar, A., Wang, J., & Chen, L. (2022). A hybrid CNN-Transformer model for autonomous power line anomaly detection. *Frontiers in Robotics and AI*, *9*, 102310.

[v] Wang, X., Li, Y., & Deng, M. (2021). Sample matching and post-clustering techniques for power cable detection. *Proceedings of the ACM on Multimedia Conference*, 303–311.

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